

SCREENING OF ANTIBACTERIAL ACTIVITY OF DIFFERENT BRANDS OF CO-TRIMOXAZOLE PAEDIATRIC SUSPENSION AVAILABLE IN THE LOCAL MARKET

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ABSTRACT

Sulfonamides are the drug of choice for number of infections like pneumonia, toxoplasmosis, nocardiosis, urinary tract infections. Sulfonamides are most commonly used in combinations such as "Trimethoprim-Sulfamethoxazole" (TMP+SMZ) or co-trimoxazole. Sixty five isolates belonging to five different species, *E. coli* (22), *S. aureus* (18), *Klebsiella* (05), *Pseudomonas* (16) and *Proteus* (04) were used for screening antibacterial activity against different brands (A-G) of Co-trimoxazole by disc diffusion method. Brand G exhibited highest activity against *E. coli* with mean zone of inhibition $41\text{mm} \pm 2.3$ standard deviation. The antibacterial activity against other species *S. aureus*, *Klebsiella* and *Pseudomonas* were found as: (mean $\pm$ standard deviation) observed as $39.73\text{ mm} \pm 10.59\text{ mm} \pm 38\text{ mm} \pm 2$ and $24.93\text{ mm} \pm 5.32$. However all isolates of *Proteus* were found resistant against different brands of co-trimoxazole.

INTRODUCTION

The sulfonamides drugs were the first effective, chemotherapeutic agents to be employed systemically for the prevention and cure of bacterial infections in human beings. The considerable medical and public health importance of their discovery and their subsequent widespread use were quickly reflected in the sharp decline in morbidity and mortality figures for the treatable infectious diseases. The advent of penicillin and subsequently of other antibiotics has diminished the usefulness of the sulfonamides and the presently occupy a relatively small place in the therapeutic armamentarium of the physician. However, the introduction in the mid 1970s of the combination of trimethoprim and sulfamethoxazole has resulted in increased use of sulfonamides for the treatment of specific microbial infections.

Except for sulfonamides especially designed for their local effects in the bowel, this class of drugs is rapidly absorbed from the

gastrointestinal tract. Approximately 70% to 100% of an oral dose is absorbed and sulfonamide can be found in the urine within 30 minutes of ingestion. Peak plasma levels are achieved in 2 to 6 hours, depending on the drug. All sulfonamides are bound in varying degree to plasma proteins particularly to albumin (William A. Petri, Jr 2001).

Sulfonamides are distributed throughout all tissues of the body and readily enter pleural, peritoneal, synovial, ocular and similar body fluids and may reach concentrations therein that are 50% to 80% of the simultaneously determined concentration in blood. After systemic administration of adequate doses, sulfadiazine and sulfisoxazole attain concentrations in cerebrospinal fluid that may be effective in meningeal infections. At steady state, the concentration ranges between 10% and 80% of that in the blood. However, because of the emergence of sulfonamide-resistant microorganisms, these drugs are now used only rarely for the treatment of meningitis (William A. Petri, Jr. 2001).

Table 1
Antibacterial activity of different brands of co-trimoxazole (Trimethoprim + sulfamethoxazole) against *E. coli*, *S. aureus*, *Klebsiella*, *Pseudomonas* and *Proteus*

Name of organisms	No. of isolates	Mean zone of inhibition in mm $\pm$ standard deviation						
		A	B	C	D	E	F	G
<i>E. coli</i>	22	37.7 $\pm$ 4.81	37.4 $\pm$ 5.04	36.33 $\pm$ 5.10	36.58 $\pm$ 5.64	39.4 $\pm$ 6.11	38.3 $\pm$ 6.03	41 $\pm$ 2.30
<i>S. aureus</i>	18	40.66 $\pm$ 12.03	38.66 $\pm$ 11.66	38.46 $\pm$ 11.22	39.13 $\pm$ 10.63	39.13 $\pm$ 11.82	39.93 $\pm$ 12.40	39.73 $\pm$ 10.59
<i>Klebsiella</i>	05	35 $\pm$ 5.00	35 $\pm$ 5.00	35.5 $\pm$ 4.5	35.5 $\pm$ 4.5	39.5 $\pm$ 0.5	39.5 $\pm$ 0.5	38 $\pm$ 2.00
<i>Pseudo-monas</i>	16	21.84 $\pm$ 7.19	22.53 $\pm$ 6.02	19.08 $\pm$ 8.04	18.83 $\pm$ 7.92	23 $\pm$ 4.83	22.57 $\pm$ 5.42	24.93 $\pm$ 5.32
<i>Proteus</i>	04	-	-	-	-	-	-	-

Trimethoprim in combination with sulfamethoxazole constitutes an important role in the development of clinically effective antimicrobial agents. This combination is also known as “Co-Trimoxazole” (Wormser *et al.*, 1982). The antibacterial spectrum of trimethoprim is similar to that of sulfamethoxazole, although the former drug is usually 20 to 100 times more potent than the latter. Most Gram negative and Gram positive microorganisms are sensitive to trimethoprim but resistance can develop when the drug is used alone (Ward *et al.*, 1982). *Pseudomonas aeruginosa*, *Bacteroides fragilis* and *Enterococci* usually are resistant. There is a significant variation in the susceptibility of Enterobacteriaceae to trimethoprim in different geographical locations because of the spread of resistance mediated by plasmids and transposons (Goldstein *et al.*, 1986).

Sulfonamides originally had a wide range of activity but this range had now been seriously restricted by acquired bacterial resistance. Trimethoprim and co-trimoxazole and other trimethoprim combinations are used these days as they have synergistic effect. Further with the advent of penicillin and subsequent other antibiotics have diminished the usefulness of sulfonamides.

Keeping in view the less exposure of sulfonamides and their combinations, the present studies have been designed.

MATERIAL AND METHODS

Equipment and Apparatus:

Autoclave, incubator, hot air oven, balance, glasswares, cork boarer, forceps, wire loop (Sterile Swabs).

Media and Chemicals:

1. Muller Hilton Agar, 2. Muller Hilton Broth

To evaluate the antibacterial activity of cotrimoxazole, a combination of Trimethoprim and sulfamethoxazole (TMP + SMZ) against different clinical isolates such as *Escherichia coli*, *Staphylococcus aureus*, *Proteus* species, *Klebsiella* species and *Pseudomonas* species, seven different brands of cotrimoxazole (TMP + SMZ) (A-G) in the paediatric suspension available in the market were evaluated by disc diffusion method.

Preparation of Media:

The Muller Hilton Agar (MHA) was prepared from dehydrated base according to the manufacturer's instructions. After autoclaving the medium is cooled to 45-50°C in a water bath. Sufficient medium to achieve a uniform depth of 4 mm is poured into Petri dishes placed on a flat surface. This will require approximately 60 and 25 mL for 14 and 9 cm plates respectively. These may be safely stored in refrigerator temperature for one week or longer if held in properly sealed containers which prevent excess loss of

moisture. The pH of the medium should be checked and should be 7.3 ± 0.1 .

PROCEDURE

The prepared plates were inoculated with the broth culture which was activated with the help of sterile wire loop. Then wells were made in the medium with sterile cork boarer and were filled with equal volume of paediatric suspension and incubated at 37°C for 24 hours. After incubation, the zone of inhibition were measured.

RESULTS AND DISCUSSION

The antibacterial activity of different brands of cotrimoxazole (combination of trimethoprim + sulfamethoxazole) have been observed against different microorganisms i.e. *Escherichia coli*, *Staphylococcus aureus*, *Pseudomonas* species, *Klebsiella*, *Proteus* species and their results are as follows:

A. Antibacterial Activity of co-trimoxazole (TMP + SMZ) against *Escherichia coli* (A-G)

After performing experiments to see the antibacterial activity of co-trimoxazole (TMP + SMZ) against "*Escherichia coli*" it was found that different brands of co-trimoxazole (A-G) showed antibacterial activity against *E. coli* which is as follows: 37.7 ± 4.81 , 37.4 ± 5.04 , 36.33 ± 5.10 , 36.58 ± 5.64 , 39.4 ± 6.11 , 38.3 ± 6.03 , 41 ± 2.30 in terms of mean zone of inhibition in mm $\pm$ standard deviation respectively (Table 1). The present results are in confirmation with that of Alonso Sanz and Abad Becquer (2003). They also reported similar results of *Escherichia coli* to Trimethoprim-Sulfamethoxazole (TMP and SMZ).

B. Antibacterial Activity of co-trimoxazole (TMP + SMZ) against *Staphylococcus aureus* (A-G)

When the antibacterial activity of co-trimoxazole (TMP+SMZ) against *Staphylococcus aureus* by using different brands of co-trimoxazole (A-G) are evaluated, the results are: 40.66 ± 12.03 , 38.66 ± 11.66 , $38.46 \pm$

11.22 , 39.13 ± 10.63 , 39.13 ± 11.82 , 39.93 ± 12.40 , 39.73 ± 10.59 in terms of mean zone of inhibition in mm $\pm$ standard deviation respectively (Table 1). The present results are in confirmation with the results of Yeldandi *et al* (1988) that trimethoprim-Sulfamethoxazole has proved to be effective against *Staphylococcus aureus* even resistant to number of other antibiotics. They further reported that this combination can be a economical and effective in the treatment of infection caused by Methicillin Resistant Strains of *Staphylococcus aureus* (MRSA). Scheld *et al.* (1987) and Agger *et al.* (2002) had also reported about the effectiveness of this combination.

C. Antibacterial Activity of Co-trimoxazole (TMP + SMZ) against *Proteus* species (A-G):

After performing experiments to see the antibacterial activity of co-trimoxazole (TMP + SMZ) against *Proteus* species, all isolates were found resistant against different brands of co-trimoxazole.

The present results is in confirmation with the report of Steinke *et al.* (2001). They had also reported increasing Trimethoprim resistance at hospital of Dundee, U.K.

D. Antibacterial Activity of Co-trimoxazole (TMP + SMZ) against *Klebsiella* (A-G):

Different brands of Co-trimoxazole (A-G) showed antibacterial activity against *Klebsiella* species as: 35 ± 5.00 , 35 ± 5.00 , 35.5 ± 4.5 , 35.5 ± 4.5 , 39.5 ± 0.5 , 39.5 ± 0.5 , 38 ± 2.00 in terms of mean zone of inhibition in mm $\pm$ standard deviation respectively (Table 1). The present results are in confirmation with the report of Metlay *et al.* (2003) at Philadelphia, USA. They reported increasing resistance to trimethoprim-sulfamethoxazole to Gram negative bacteria.

E. Antibacterial activity of co-trimoxazole (TMP + SMZ) against *Pseudomonas* species (A-G):

When the antibacterial activity of co-trimoxazole (TMP + SMZ) against *Pseudomonas* species are evaluated by using different

brands of co-trimoxazole (A-G) and the results are: 21.84 ± 7.19 , 22.53 ± 6.02 , 19.08 ± 8.04 , 18.83 ± 7.92 , 23 ± 4.83 , 22.57 ± 5.42 , 24.93 ± 5.32 in terms of mean zone of inhibition in mm $\pm$ standard deviation respectively (Table 1). The present results are in confirmation with the work of Fass *et al.* (1996). They obtained that this combination is effective against *Pseudomonas* species.

CONCLUSIONS

It can be concluded that presently sulfonamides and its combinations are still very effective against different clinical isolates. So the present work will be helpful for the pharmacist and doctors in prescribing the antimicrobials.

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